

Roll No. :

END SEMESTER EXAMINATION DECEMBER 2025

Name of the Program: **B. Tech (AI & DS)**

Semester: **III**

Course Name: **Introduction to Statistical Data Science**

Course Code: **TCS 342**

Time: **3 Hours**

Maximum Marks: **100**

Note:

- i. All questions are compulsory
- ii. Answer any Two sub-question from each main question
- iii. Total marks for each sub-question is 10
- iv. Write clearly any assumption made in problem solving

Q1		(20 Marks)	
(a)	Define the following terms w.r.t data science and cite <i>two</i> examples for each: Data, Information, Knowledge, Language, Wisdom and Philosophy		CO 2
(b)	What is big data source and how does it differ from normal data set		CO 3
(c)	By citing at least one example, describe the following terms in data science: (a) Fundamental theorem of Probability, (b) Random variable, (c) Bayes Theorem, (d) Data		CO 2
Q2		(20 Marks)	
(a)	Formulate closed form expressions of <i>mass / density</i> function, <i>distribution</i> function, <i>mean</i> and <i>variance</i> for Bernoulli, Poisson, Gaussian, Exponential random variables.		CO3
(b)	State and derive Markov bound. Explain the Central Limit Theorem, discuss its utility for ensemble estimate such as – Sample Mean, Mean Square Error.		CO2
(c)	State Kolmogorov axioms for <i>Probability measure</i> and show that following mass / density functions satisfy Kolmogorov conditions: (i) Binomial (ii) Uniform		CO3
Q3		(20 Marks)	
(a)	What is memoryless property of random variable, prove that Geometric and Exponential random variables exhibit this property.		CO1
(b)	State and derive the Bayes Theorem. Define Probability as measurable function.		CO2
(c)	Given joint density $f(x, y) = k$, $0 < x < y < 1$; obtain: $E(x)$, $E(y)$, $Cov(x, y)$, $f(X/y)$		CO2
Q4		(20 Marks)	
(a)	Derive Linear regression model for 2- dimensional variable space. Does inclusion of additive noise improve fitness of linear regression.		CO3
(b)	Describe the following (may use figures, Math, text and examples): (a) Probability vs. Statistics, (b) ANOVA		CO2
(c)	Two originating Bus stops U 1 and U 2 have bus arrivals that are each Poisson distributed with mean 20 and 30 respectively. Assume both bus stops are equiprobably chosen then, determine: $Pr(10 \text{ arrivals} / U1)$ and, $Pr(15 \text{ arrivals})$.		CO3

Q5	(20 Marks)	
(a)	$\underline{X} = \{ 2.2, 3.2, 4.7, 23, 25, 44, 58, 79, 81, 109 \}$, and $\underline{Y} = \{ 44, 42, 45, 56, 67, 89, 99, 129, 189, 211 \}$ are arriving features for 10 tuple space. Compute Person correlation and linear regression co-efficient for the dataset.	CO2
(b)	State the role of Eigen decomposition theorem in generating Principal components. Give properties of Eigen representation for covariance matrix. For covariance matrix of 3 tuple space $\underline{X} = [X_1, X_2, X_3]$ below compute eigen-representation as PCA. $\underline{K}_{(X,X)} = \begin{bmatrix} 4 & 6 & 11 \\ 6 & 9 & 2 \\ 11 & 2 & 6 \end{bmatrix}$	CO3
(c)	Given 2 cubic dice are denoted by X and Y, say Z can assume the random variables: (i) $X - Y$, (ii) $X + Y$, (iii) $4X$, (iv) $2Y + 3$. Obtain true Mean, Variance and comment on what do these statistics cover in all <i>four</i> cases.	CO2